For Examiner's Use

5 A stationary copper-64 nucleus undergoes a nuclear reaction to form a zinc-64 nucleus and a beta-particle. After the reaction, the zinc-64 nucleus moves off with negligible speed, while the beta-particle moves off with high speed.

(a) Explain qualitatively why the beta-particle moves off with a much higher speed than the zinc-64 nucleus.

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..... [2]

(b) The kinetic energy E_K and the distance r of the beta-particle from the centre of nucleus of zinc-64 nucleus' centre are monitored. Fig. 5.2 shows how the values of E_K varies with $\frac{1}{r}$.

Assume that the beta-particle and zinc-64 nucleus form an isolated system.

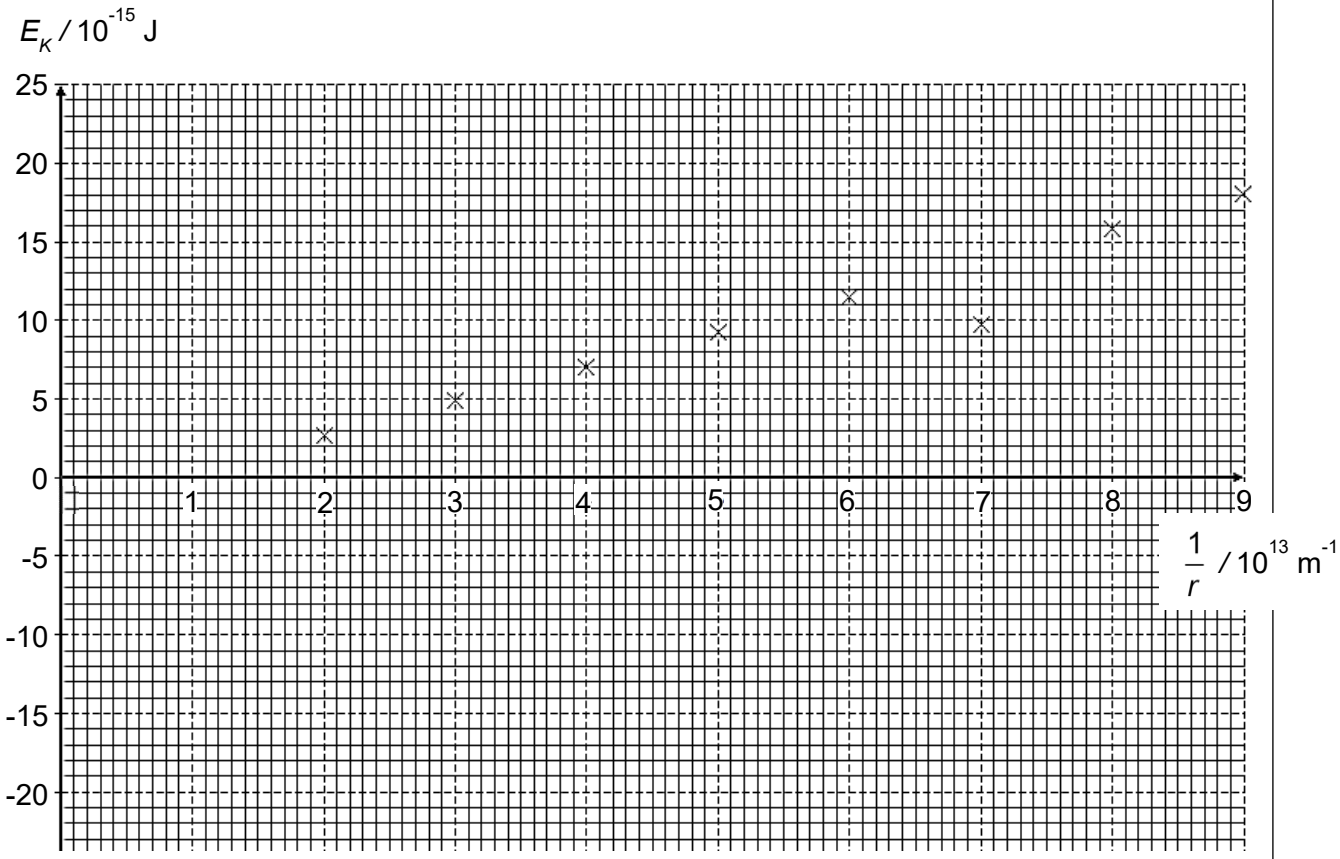

Fig. 5.2

(i) By drawing an appropriate line on Fig. 5.2, determine the y-intercept, E_o, of the graph.

vertical intercept, E_o = ... J [2]

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(ii) Explain the physical significance of the value of E_o found in **(b)(i)**.

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…...…….. [2]

(iii) Assuming that total energy of the beta-particle remains constant during its motion, sketch labelled graphs on the axes in Fig. 5.2 to represent

1. the variation of the total energy E_T .

2. the variation of the electric potential energy E_P. [3]